

Rishi Premhani

B.Tech in Electronics and Communication Engineering
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Research Interests

Medical Image Analysis • Computer Vision • Deep Learning • Representation Learning • Diffusion Models • Variational Autoencoders • Capsule Networks

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat Expected Graduation : 2028
B.Tech in Electronics and Communication Engineering CGPA: 7.40

Research Experience

Undergraduate Research Assistant May 2026 – Present
Department of Electronics and Communication Engineering, SVNIT Surat Supervisor: Dr. K. P. Upla

- Working on deep learning methods for medical image enhancement and computer vision applications.
- Designing cross-modality knowledge distillation frameworks for Wireless Capsule Endoscopy (WCE) super-resolution.
- Developing latent representation learning methods using Variational Autoencoders and Latent Diffusion Models.
- Conducting ablation studies, loss-function analysis and experimental evaluation on the Kvasir dataset.

Projects

Cross-Modal Latent Anatomical Prior Distillation for Wireless Capsule Endoscopy Super-Resolution
Ongoing

- Developing a cross-modality dual-teacher knowledge distillation framework to transfer anatomical priors from conventional endoscopy images to Wireless Capsule Endoscopy (WCE).
- Designing a latent-space representation learning pipeline using Variational Autoencoders for anatomy-preserving feature extraction.
- Learning modality-invariant representations to improve reconstruction fidelity while preserving clinically relevant structures.
- Experimental evaluation on the Kvasir dataset.

Unsupervised Latent Diffusion Framework for Medical Image Super-Resolution

Ablation Studies in Progress

- Developed a latent diffusion pipeline for medical image super-resolution using a pretrained Variational Autoencoder.
- Implemented latent-space denoising with UNet diffusion while preserving anatomical structures through reconstruction-based supervision.
- Integrated reconstruction, perceptual, wavelet and adversarial objectives for high-fidelity latent representation learning.
- Currently performing extensive ablation studies and loss-function analysis on the Kvasir dataset.

STM32-Based Flight Controller for Autonomous Quadcopter

- Designed and implemented a custom STM32 flight controller integrating IMU sensors and PID-based stabilization.
- Developed embedded firmware for sensor fusion, attitude estimation and real-time motor control.
- Built as a foundation for future autonomous navigation and computer vision integration.

Technical Skills

Programming: Python, C/C++, MATLAB

Frameworks: PyTorch, Torchvision, OpenCV

Machine Learning: Representation Learning, Variational Autoencoders, Diffusion Models, Knowledge Distillation

Computer Vision: Medical Image Analysis, Image Super-Resolution, Semantic Segmentation

Embedded Systems: STM32, Sensor Fusion, PID Control

Tools: Git, GitHub, Linux, VS Code, Overleaf

AI Tools: ChatGPT, Claude, Gemini, Cursor , Ollama , Local Models

Leadership

Team Leader, Business Development

Aug 2025 – July 2026

AIIESEC in Surat

- Worked with the Business Development team to establish partnerships and support organizational initiatives.
- Contributed to outreach, stakeholder communication and execution of local chapter events.
- Collaborated with cross-functional teams to deliver events and member engagement activities.